

Math 514, F19
Quiz 3

Name: Answer

Instructions: You must show all of your work to receive credits for correct answers.

Problem 1. (4 points) For a standard normal random variable Z , find $P(|Z| < 1.66)$.

$$\begin{aligned} P\{|Z| < 1.66\} &= P\{-1.66 < Z < 1.66\} \\ &= P\{Z < 1.66\} - P\{Z \leq -1.66\} \\ &= \Phi(1.66) - P\{Z \geq 1.66\} \\ &= \Phi(1.66) - (1 - P\{Z \leq 1.66\}) \\ &= \Phi(1.66) - (1 - \Phi(1.66)) \\ &= 2\Phi(1.66) - 1 \approx 2 \times 0.9515 - 1 = 0.903 \end{aligned}$$

Problem 2. (6 points) Suppose that the amount of time that a certain battery functions is a normal random variable with mean 400 hours and standard deviation 40 hours. Suppose that an individual owns two such batteries. Let X_1 and X_2 denote the lives of the first and second batteries respectively.

(a) Let $X = X_1 + X_2$ denote the sum of lives of the two batteries, which is still a normal random variable. What are the mean and standard deviation of X ?

$$\begin{aligned} X_1 &\sim N(\underbrace{400}_{\mu_1}, \underbrace{40^2}_{\sigma_1^2}), \quad X_2 \sim N(\underbrace{400}_{\mu_2}, \underbrace{40^2}_{\sigma_2^2}) \\ \text{Thus } X &\sim N(\mu, \sigma^2) \text{ with} \\ \mu &= \mu_1 + \mu_2 = 400 + 400 = 800 \\ \sigma^2 &= \sigma_1^2 + \sigma_2^2 = 40^2 + 40^2 = 3200 \Rightarrow \sigma = \sqrt{3200} = 40\sqrt{2} \end{aligned}$$

(b) What is the probability that the sum of lives of the two batteries will exceed 750 hours?

$$\begin{aligned} P\{X > 750\} &= P\left\{Z > \frac{750 - \mu}{\sigma}\right\} \\ &= P\left\{Z > \frac{750 - 800}{40\sqrt{2}}\right\} \\ &\approx P\{Z > -0.8839\} \\ &= P\{Z < 0.8839\} \\ &= \Phi(0.8839) \approx 0.8116 \end{aligned}$$